

COMPASS_SynopticCB_PW_DOC_202605_Data

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##Setup - Change things here & write any notes

#identify section

cat("Setup Information")

Setup Information

Run information - PLEASE CHANGE

Date_Run = "05/21/26" *#Date that instrument was run*

Run_by = "Stephanie J. Wilson" *#Instrument user*

Script_run_by = "Stephanie J. Wilson" *#Code user*

run_notes = "Ran a second standard curve, but the vial #'s got mixed up so not included here." *#any*

samples <- c("GCW", "GWI", "MSM", "SWH") *#whatever identifies your samples within the same names*

samples_pattern <- paste(samples, collapse = "|")

#samples_pattern <- "GCW" #use this instead of the line above if you have only one site code

chks_name = "Chk_Std_50C_2N" *#what did you name your check standards?*

File Names - PLEASE CHANGE

#file path and name for raw summary data file

raw_file_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202605.txt"

#file path and name for raw all peaks file

raw_allpeaks_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202605_allpeaks.txt"

#file path and name of processed data file

processed_file_name = "Processed Data/COMPASS_SynopticCB_PW_Processed_DOC_202605.csv"

Log Files - PLEASE CHECK

#downloaded metadata csv - downloaded from Google drive as csv for this year

Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2026.csv"

#qac log file path for this year

Log_path = "Raw Data/COMPASS_Synoptic_TOCTN_QAClog_2026.csv"

##Set Up Code

##Read in metadata and create similar sample IDs for matching to samples

Import Data Functions

Import Sample Data

```
## Import Sample Data

## New names:
## * ' ' -> '...14'

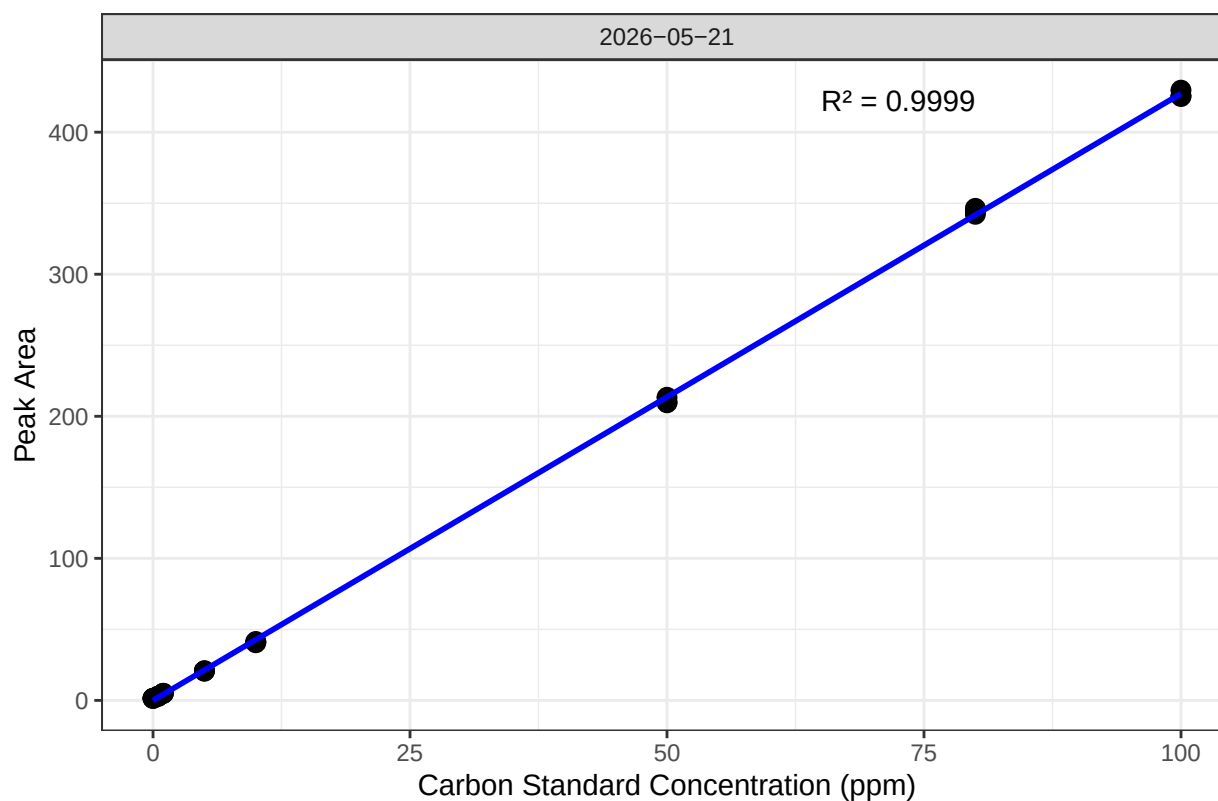
## # A tibble: 6 x 4
##   sample_name      npoc_raw tdn_raw run_datetime
##   <chr>          <dbl>   <dbl> <chr>
## 1 202605_GCW_TR_LysA_10cm      3.07  0.0729 5/22/2026 12:09:29 AM
## 2 202605_GCW_TR_LysA_20cm      2.31  0.0864 5/22/2026 12:34:30 AM
## 3 202605_GCW_TR_LysA_20cm_dup    2.36  0.158  5/22/2026 1:02:20 AM
## 4 202605_GCW_TR_LysB_10cm      4.00  0.206  5/22/2026 1:30:22 AM
## 5 202605_GCW_TR_LysB_20cm      3.13  0.382  5/22/2026 1:58:02 AM
## 6 202605_GCW_TR_LysC_10cm     15.0  0.528  5/22/2026 2:27:52 AM
```

Assessing Standard Curves

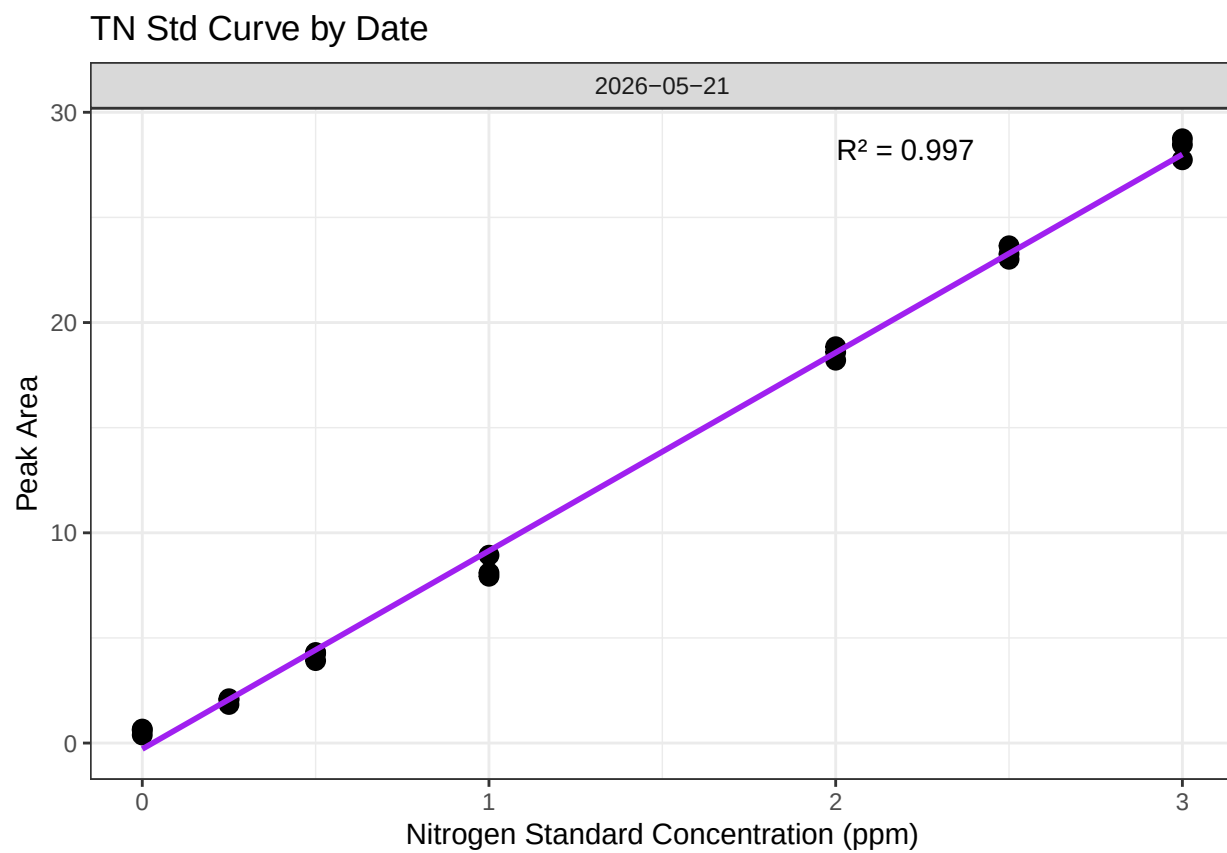
```
## Assess the Standard Curves

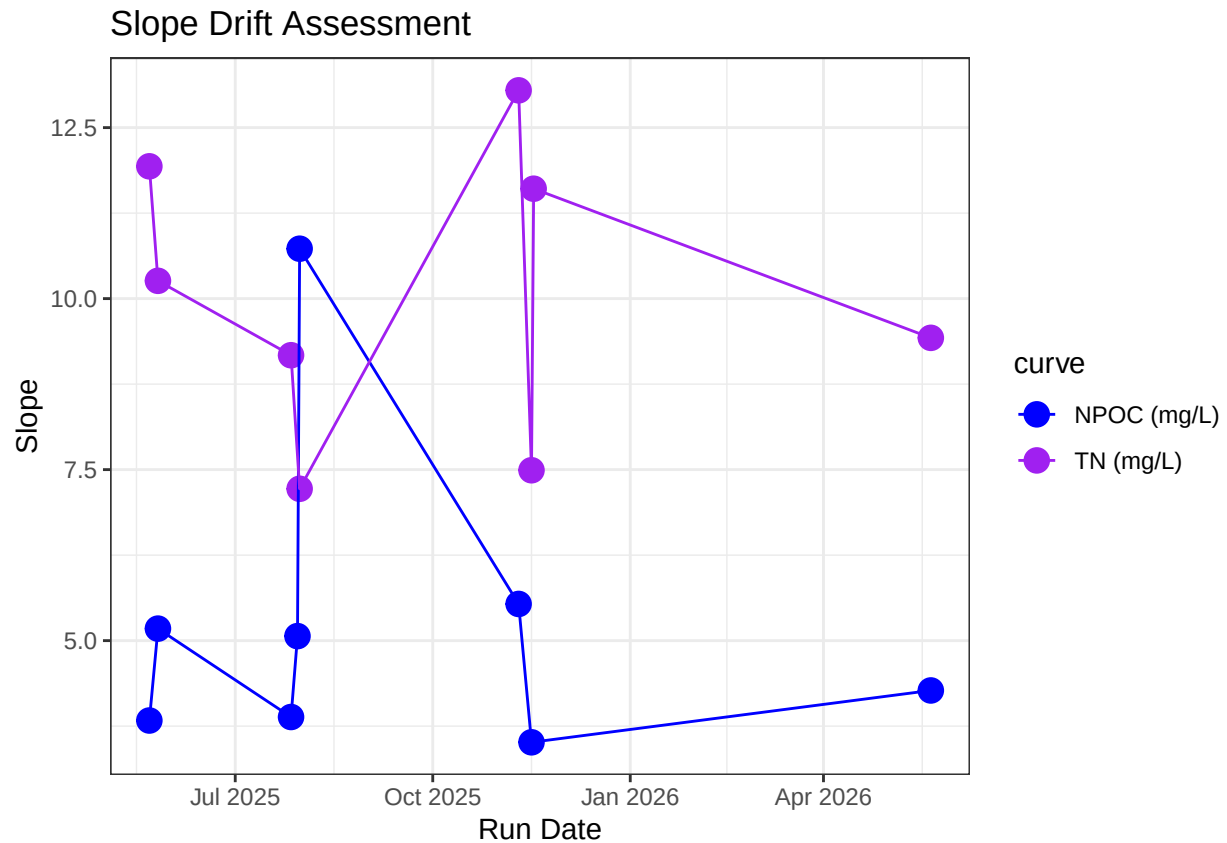
## New names:
## 'geom_smooth()' using formula = 'y ~ x'
## * ' ' -> '...18'
```

NPOC Std Curve by Date



```
## 'geom_smooth()' using formula = 'y ~ x'
```





```
## [1] "NPOC Curve r2 GOOD"
```

```
## [1] "TN Curve r2 GOOD"
```

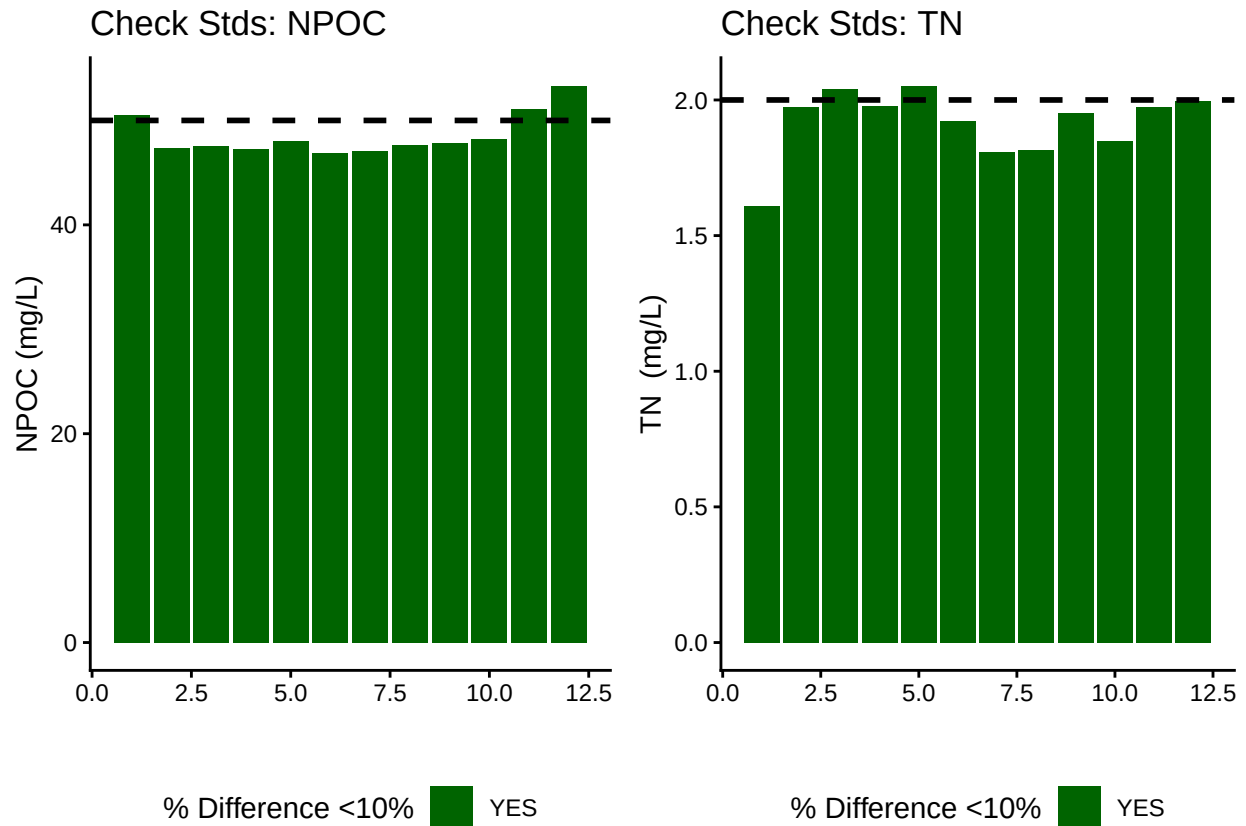
Assess Check Standards

```
## Assess the Check Standards
```

```
## New names:
## * ' ' -> '...14'
```

```
## [1] "Carbon Check Standard RSD within Range"
```

```
## [1] "Nitrogen Check Standard RSD within Range"
```



```
## [1] ">60% of Carbon Check Standards are within range of expected concentration"
```

```
## [1] ">60% of Nitrogen Check Standards are within range of expected concentration"
```

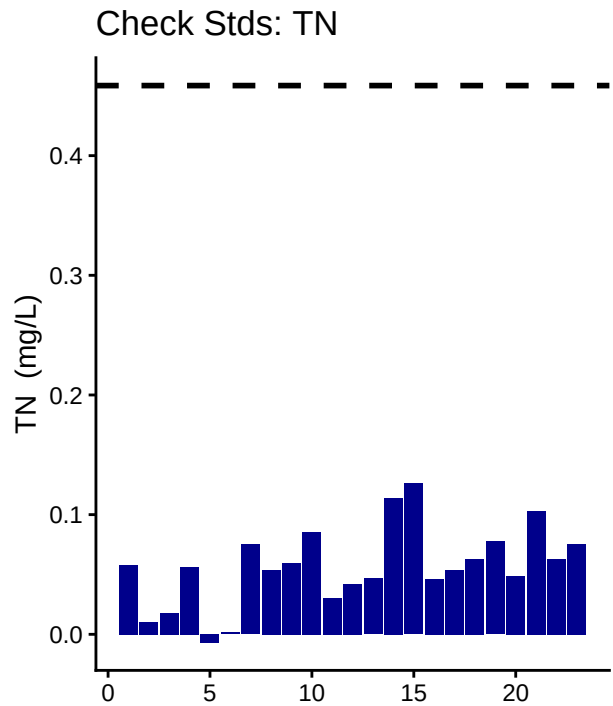
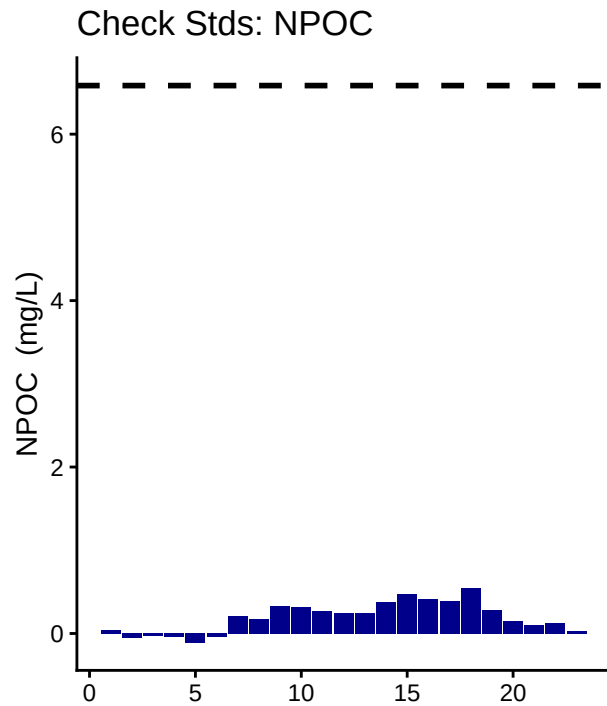
Assess Blanks

```
## Assess Blanks
```

```
## New names:
## * ' ' -> '...14'
```

```
## [1] ">60% of Carbon Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">60% of Nitrogen Blank concentrations are lower 25% quartile of samples"
```



Blank Conc <25% Quartile Samples ■ YE

Blank Conc <25% Quartile Samples ■ YE

```
## carbon blanks:
```

```
## [1] 0.1934409
```

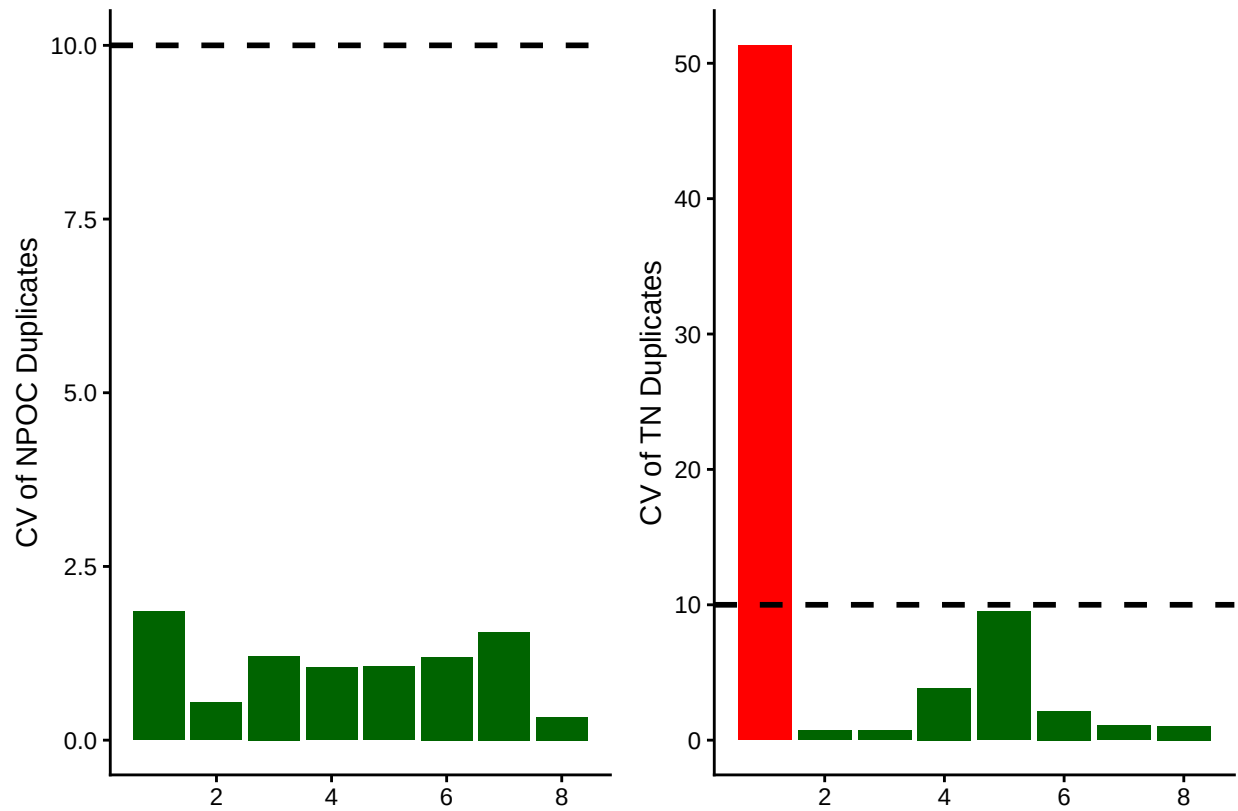
```
## nitrogen blanks:
```

```
## [1] 0.0565787
```

Assess Duplicates

```
## Assess Duplicates
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

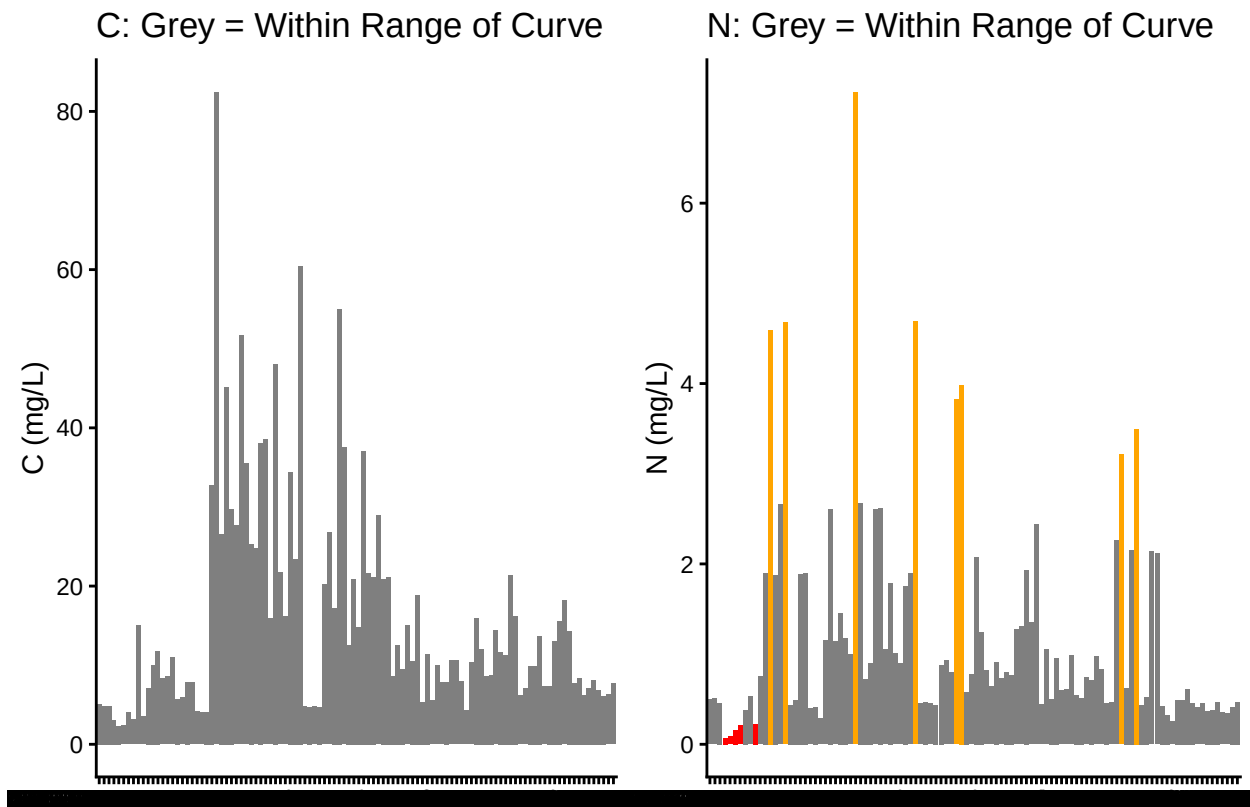


```
## [1] ">60% of Carbon Duplicates have a CV <10%"
```

```
## [1] ">60% of Nitrogen Duplicates have a CV <10%"
```

Sample Flagging - Are samples Within the range of the curve?

```
## Sample Flagging
```

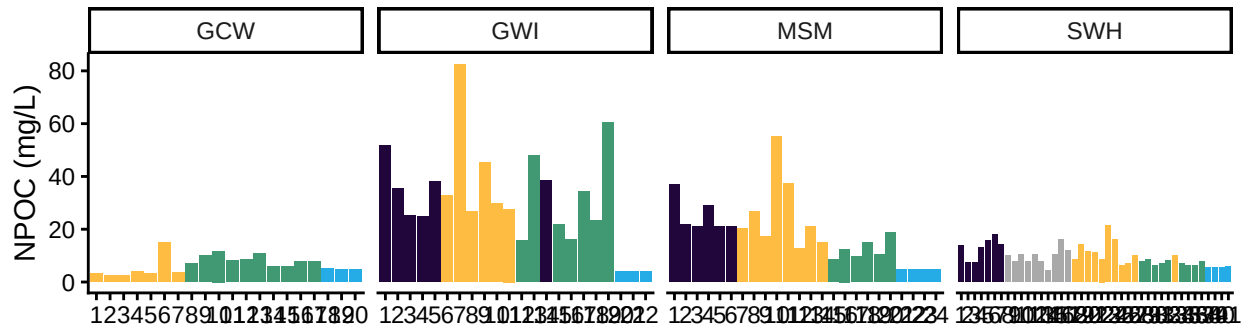


Visualize Data by Plot

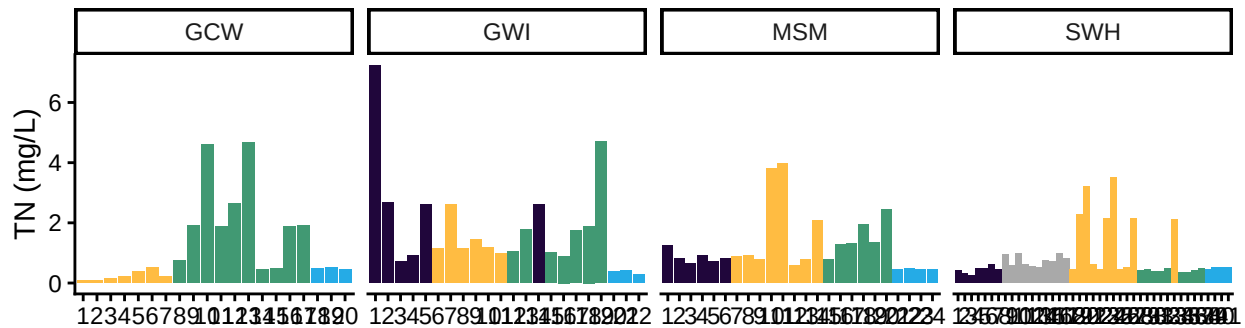
```
## Visualize Data
```

```
## Warning in rbind(c("202605", "GCW", "TR", "LysA", "10cm"), c("202605", "GCW", :
## number of columns of result is not a multiple of vector length (arg 1)
```

Samples: NPOC



Samples: TN



Convert data from mg/L to uMoles/L

Check to see if samples run match metadata & merge info

```
## Check Sample IDs with Metadata
```

```
## All sample IDs are present in metadata.
```

Export Processed Data

```
## Export Processed Data
```

```
## # A tibble: 6 x 21
```

```
##   Project      Region Site Zone Replicate Depth_cm Sample_ID Year Month Day
##   <chr>        <chr> <chr> <chr> <chr>      <int> <chr>      <int> <int> <int>
## 1 COMPASS: Sy~ CB    GCW  TR   A          10 202605_G~  2026    5   12
## 2 COMPASS: Sy~ CB    GCW  TR   A          20 202605_G~  2026    5   12
## 3 COMPASS: Sy~ CB    GCW  TR   B          10 202605_G~  2026    5   12
## 4 COMPASS: Sy~ CB    GCW  TR   B          20 202605_G~  2026    5   12
## 5 COMPASS: Sy~ CB    GCW  TR   C          10 202605_G~  2026    5   12
## 6 COMPASS: Sy~ CB    GCW  TR   C          20 202605_G~  2026    5   12
## # i 11 more variables: Time <chr>, Time_Zone <chr>, npoc_mgL <dbl>,
## #   npoc_uM <dbl>, npoc_flag <chr>, tdn_mgL <dbl>, tdn_uM <dbl>,
## #   tdn_flag <chr>, Analysis_runtime <chr>, Run_notes <chr>, Field_notes <lgl>
```

#end